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M 008 VERTICAL SUMP PUMPS

1 SCOPE OF WORK

The vertical sump pumps specified in **Part 4, Subsection 1.2 "Summary of Work", in Section 3 Technical Specifications** or as required by the Bidder's design and the required electrical equipment specified in **Clause 8** herein, shall be subject to the design, testing and quality assurance requirements of this Specification as applicable.

2 CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 "Codes & Standards"**. The Codes and Standards of the following organizations extracted from **Subsection 2.5** or international equivalent standards are specifically applicable to the design manufacture and testing of the equipment included in this specification.

AISI	American Iron and Steel Institute
ANSI	American national Standards Institute
ASME	American Society of Mechanical Engineers
ASTM	American Society of Testing and Materials
AWWA	American Water Works Association
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HIS	Hydraulic Institute Standards
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronic Engineers
JIS	Japan Industrial Standard

3 DESIGN AND OPERATING CONDITIONS

3.1 General

Sump pumps with discharge piping shall be provided wherever necessary to dewater any pits, trenches or sumps which are required to accommodate the Bidder's design.

Pumps shall be of the submersible type and arranged for easy withdrawal for maintenance. They shall discharge into a convenient floor drain and shall be equipped with float switches for automatic operation.

The pump capacities shall be such that pits or trenches shall be emptied from full in a reasonable time appropriate to the application. In no case shall this time be greater than two hours.

3.2 Steam Turbine Oil Sump Pump

Description	Requirement
Equipment Name	Steam Turbine Oil Tank Sump Pump
Quantity	2 x 100%
Location	Turbine Building
Liquid to be handled	Drains from the Steam Turbine Oil tanks
Temperature of liquid (°C)	20 - 40
Pump suction from	Drain sump
Pump discharge to	Oil Separator
Pump capacity each (l/s)	By the Bidder
External discharge head (does not include pump column friction losses) (m)	By the Bidder
Net developed head (m)	By the Bidder
Material	Manufacture's standard suitable for the intended service, subject to the owner's acceptance
Service	Intermittent

		Description	Requirement
		Type of design	Submerged type
		Pump service speed (rpm)	Manufacture's standard suitable for the intended service, subject to the owner's acceptance
3.3	Gas Turbine Area (if Applicable)		
		Description	Requirement
		Equipment Name	Gas Turbine Oil Tank Sump Pump
		Quantity	2 x 100%
		Location	Gas Turbine Area
		Liquid to be handled	Drains from the Gas Turbine Oil tanks
		Temperature of liquid (°C)	20 - 40
		Pump suction from	Drain sump
		Pump discharge to	Oil Separator
		Pump capacity each (l/s)	By the Bidder
		External discharge head (does not include pump column friction losses) (m)	By the Bidder
		Net developed head (m)	By the Bidder
		Material	Manufacture's standard suitable for the intended service, subject to the owner's acceptance
		Service	Intermittent
		Type of design	Submerged type
		Pump service speed (r/m)	Manufacture's standard suitable for the intended service, subject to the owner's acceptance
3.4	Condenser Sump Pump		
		Description	Requirement
		Equipment Name	Condenser Pit Sump Pump
		Quantity	2 x 100%

Description	Requirement
Location	Condenser pit area
Liquid to be handled	Seawater
Temperature of liquid (°C)	40
Pump suction from	Drain sump
Pump discharge to	Floor drain
Pump capacity each (l/s)	By the Bidder
External discharge head (does not include pump column friction losses) (m)	By the Bidder
Net developed head (m)	By the Bidder
Material	Suitable for very corrosive service
Service	Intermittent
Type of design	Submerged type
Pump service speed (rpm)	Manufacture's standard suitable for the intended service, subject to the owner's acceptance

3.5 HRSG Area Sump Pump

Description	Requirement
Equipment Name	HRSG Area Pit Sump Pump
Quantity	1 x 100%
Location	HRSG
Liquid to be handled	Water/Oil
Temperature of liquid (°C)	20 - 40
Pump suction from	Elevator Pit sump
Pump discharge to	Oil Separator
Pump capacity each (l/s)	By the Bidder
External discharge head (does not include pump column friction losses) (m)	By the Bidder
Net developed head (m)	By the Bidder
Material	Suitable for very corrosive service

	Description	Requirement
	Service	Intermittent
	Type of design	Submerged type
	Pump service speed (rpm)	Manufacture's standard suitable for the intended service, subject to the owner's acceptance

3.6 CWP Area Sump Pump

	Description	Requirement
	Equipment Name	CWP Sump Pump
	Quantity	1 x 100%
	Location	CWP intake
	Liquid to be handled	Seawater
	Temperature of liquid (°C)	20 - 40
	Pump suction from	CWP Pit
	Pump discharge to	Trash Pit
	Pump capacity each (l/s)	By the Bidder
	External discharge head (does not include pump column friction losses) (m)	By the Bidder
	Net developed head (m)	By the Bidder
	Material	Suitable for very corrosive service
	Service	Intermittent
	Type of design	Submerged type
	Pump service speed (rpm)	Manufacture's standard suitable for the intended service, subject to the owner's acceptance

4 DETAILED REQUIREMENTS

4.1 General

Each pump and its appurtenances and accessories shall conform to the requirements of this Specification.

The Total Head (h) shall be calculated based on commercial pipes (not new and clean) as indicated in Williams & Hazen's Formula.

4.2 Pump Type and Design

Unless otherwise specified, pump(s) shall be set in a concrete sump pit with the motors mounted at the floor level above the pit. An opening shall be provided in the concrete floor over the pit with suitable curb angles around the opening to accommodate the sump cover plate. The sump cover plate shall be furnished with the pump. The design pump rpm shall not exceed 900 rpm.

A "Duplex" pump unit arrangement shall consist of two pumps mounted on a common sump cover plate and installed in a common sump pit. Sump cover plate design shall be such that each individual pump can be removed as a unit without disturbing the operation of the remaining pump.

A single pump unit arrangement, where specified, shall consist of one pump mounted on its individual sump cover plate.

Each pump shall be of the vertical shaft, submerged or cantilevered type as indicated in Clause 4.10 herein, centrifugal, non-clogging type, supported from its motor base plate by means of a suspension pipe. The pump shall either be direct-connected to its motor by means of a suitable dynamically-balanced flexible coupling or connected to its motor by means of a belt drive resulting in a self contained unit that can be lifted from the pit. All parts of the equipment shall be readily accessible for inspection, adjustment and repairs.

Each pump shall be capable of operation from minimum flow to run-out without cavitation or undue vibration and distress. Run-out shall be interpreted as a pump's maximum flow condition when in service individually with no other pump in operation.

Each pump, unless it is of the special solids-passing type, shall be equipped with a cast iron suction screen secured in place with corrosion-resistant bolting. However, the material selected by the Bidder for low pH service (waste water) shall be entirely suitable for the service required.

Pumps shall be designed and constructed to withstand the most severe service conditions under which the pump will be required to operate. All materials

of construction shall be identified with ASTM standard numbers. All features of design and construction and all materials shall be acceptable to the Owner.

Pumps shall be specifically designed to suit the operating conditions anticipated and as outlined in Clause 3 herein for each particular case. The design and construction of the pumps and appurtenances shall be such that they will operate over a long period of years with a minimum of wear, erosion or corrosion of any operating parts. The parts subject to unavoidable wear, erosion or corrosion, shall be the best material available for the purpose and shall be acceptable to the Owner.

Bidder shall be responsible for determining that the pump and motor operate in dynamic balance as a unit without undue vibration or noise. The vibration levels as measured on the pump thrust bearing housing or the top motor bearing, if the thrust bearing is located in-the motor, shall not exceed the level specified by the Hydraulic Institute at design speed and flow conditions. This same vibration level shall also not be exceeded when these pumps are coming up to speed or when operating at less than full flow. Bidder shall verify that the critical speed of the complete pump and motor unit is at least 25% above the motor operating speed (rpm).

Lifting lugs or eyes shall be provided as required to facilitate handling.

4.3 Pump Characteristics

Each pump shall have a stable, steadily rising head characteristic curve with decrease in capacity, suitable for the service specified. Each pump and motor combination shall have non overloading characteristics.

Duplicate pumps designed for the same conditions shall operate satisfactory in parallel with each other or with other pumps of similar characteristics.

The pumps characteristics shall include NPSH required (NPSHR) curves based on "as tested" results based on a 0% head deviation at the "break away" points. The recommended minimum NPSHR for normal continuous operation shall be indicated in the curves, for all flows including maximum run-out. The length of the pumps and minimum submergence shall be sufficient to suppress cavitation including the recommended margin and prevent vortexing of the liquid entering the pump.

A minimum margin of 25% shall be maintained between NPSHR and NPSHA.

4.4 Casing and Suspension Pipe

Pump casing shall be fine grained cast iron for general service. Special Materials subject to review by the Owner, shall be provided for abrasive and/or corrosive service. Pump casing shall be rigidly secured to a heavy-wrought steel suspension pipe. The suspension pipe shall be of material that is of equivalent corrosion-resistance to and compatible with the pump casing material.

Joints in pump suspension pipe shall be flanged, male and female, tongue and grove type, or other similar approved construction, to ensure proper fit and alignment.

4.5 Shaft and Impeller

Pump shaft shall be of suitable type steel and of ample diameter to prevent any "whipping" action or other disturbance under any operating conditions. Renewable, grease or oil lubricated sleeve guide bearings shall be provided when required.

Impellers shall be of the non-clogging type, made of cast iron for general service. Special materials, subject to review by the Owner, shall be provided for abrasive and/or corrosive service. Impellers shall be accurately machined, smoothly finished, tightly fitted and rigidly keyed and locked to the pump shaft. Impellers shall be statically and hydraulic-balanced radially to ensure smooth operation under all conditions.

4.6 Bearings

Bearings shall be of design with ample wearing surfaces and equipped with a forced grease feed system or with sight feed lubrication for general service. Grease fittings or oil cups shall be conveniently located above the baseplate. For abrasive service where the bearings are located below the baseplate, the bearings shall be of the cutlass rubber type lubricated with clean flushing water.

Intermediate guide bearings shall be provided, where necessary, so that the distance between bearings will not exceed 1500 mm.

Thrust bearings shall be designed to handle the hydraulic and mechanical load with ample safety factor.

Thrust bearing may be located in the driving motor or in a scaled housing above the floor plate. If the thrust bearing is in the motor, a dynamically balanced rigid adjustable coupling shall be furnished.

If a belt drive is furnished, the bearings shall be sized to carry the loads on the shaft from the belt drive in addition to the normal design loads.

4.7 Packing Glands

Packing glands materials selected for abrasive and/or corrosive service shall be subject to review by the Owner.

4.8 Sump Cover Plate

Sump cover plate shall not be less than 13 mm thick. Cover plate shall be of heavy construction, made of steel plate unless otherwise specified and provided with a screened vent pipe and a manhole. Manhole shall be approximately 300 mm x 400 mm, 400 mm diameter, or manufacturer's nearest standard size. Should plate consist of two or more parts, the latter shall be flanged and bolted together.

4.9 Discharge Pipe

Pump discharge pipe shall be steel for general service. Pipe shall terminate just above the baseplate with a Class 125 or Class 150 ANSI standard pipe flange if pipe is 64 mm or larger in size and with a threaded outlet if pipe is 50 mm or smaller in size. Connection of the discharge pipe to the pump shall be designed to allow removal of the entire pump unit for maintenance and repair without dismantling any portion of the discharge pipe. Special materials, subject to review by the Owner, shall be provided for abrasive and/or corrosive service.

4.10 Cantilever Sump Pump

The cantilever sump pumps shall be designed and constructed in accordance with the applicable requirements of this specification with exception and/or additions stated below:

- The pumps shall be of a cantilever design with no submerged bearings. Suction tail pipes may be utilized if required by Bidder.
- Hydraulic balancing devices shall not be utilized.
- The first critical speed shall exceed the maximum operating speed by at least 25%.
- All joints located below the cover plate shall be of through bolt construction.
- Throttle bushings of close tolerances below the cover plate are acceptable.
- Replaceable wearing rings or surfaces are not acceptable. The pump casing shall include increased metal thickness at the points of greatest wear.
- Bidder shall state that the sump pump materials are suitable for the specified services. Materials shall be acceptable to the Owner.

4.11 Painting

Before shipment, all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and all exterior-ferrous surfaces shall be painted in accordance with the requirement in **Part 4 Section 3, S-420 and S-430 Cleanliness and Painting**.

4.12 Noise

Bidder shall meet the requirements specification **Part 4, Subsection 2.9**.

5 PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Test

All pumps shall be shop tested in accordance with Hydraulic Institute Standards, unless the Bidder can show that similar size and type of pump have been tested and the test results are applicable to the pumps offered.

5.2 Site Test

After equipment is put in service, Bidder shall run a check test on the equipment to ascertain that it is performing properly in every respect. Testing shall be in accordance with the guidelines and requirements contained in the Hydraulic Institute Standards. Any deficiencies brought out by this test shall be promptly corrected by the Bidder at no cost to Owner.

6 QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance shall conform to the requirements of **Part 4, Subsection 2.7**.

7 INSTRUMENTATION AND CONTROLS

Technical requirements governing the design and supply of instrumentation and control devices shall be as detailed in **Part 4 Section, 3 I-001** Instrumentation and Control.

Any special instrumentation and control devices integral with the pumps and appurtenance shall be supplied and installed by the Bidder. The Bidder shall include details of any such devices in his Proposal.

The Bidder shall furnish necessary float switches for complete automatic operation of the pump for the sumps. Level switch control shall be arranged so that when the water rises to a predetermined level, pump shall start automatically and continue to run until the water level drops to a predetermined level. An alarm shall be provided on the control panel which will be actuated when the water level continues to rise.

All controls mounted at the pumps shall be easily accessible for inspection and maintenance.

If pumps are furnished with bearings arranged for lubrication by an external source of clean water, solenoid-operated automatic water supply valves and pressure switches to delay start of the pumps shall be furnished until lubricating water pressure has been established. Interlocks shall be provided with the pump control system.

8 ELECTRICAL SPECIFICATION

All motors shall be in accordance with **Part 4 Section3, E-210** low voltage motors.

The pump motor shall be capable of stopping and starting a minimum of eight times per hour.

Driving motor in each case will be of the full-voltage starting type and all parts of the pump unit shall be designed to with stand the resulting stresses.

Pump and motor design shall be such that no damage will result from reserve flow due to check valve failure in discharge line.

Electrical accessories shall be in compliance **with Part 4 General Electrical Specification**. The minimum degree of protection shall be IP 66 as defined by IEC 529.

All pumps combination starters shall be located in the station motor control centers.

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